

JobsXperts – Mock & Live Test Platform (MVP)

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Scope: Multi-tenant, subdomain-based test platform with SSO, supporting Mock and Live exams with MCQ/True-False questions, scheduling, attempt engine, scoring, and result publishing.

1. Objectives

- Provide a robust Mock and Live exam platform for institutes and teachers.
- Support per-institute subdomains (`{institute-slug}.jobsxperts.com`) with SSO from JobsXperts.
- Enable scheduling (fixed-time and windowed), secure attempt flow, autosave, resume, and fair scoring.
- Offer instant results for Mock; controlled publish for Live; global and institute-wise rankings.

2. Roles & Access

- **Admin:** Global control, manage tenants, policies, and platform settings.
- **Sub-Admin:** Delegated admin operations.
- **Institute:** Organization account; manages teachers, groups, tests.
- **Teacher:** Creates question bank and tests; schedules and reviews results.
- **Student:** Takes tests (mock/live), views results and answer summaries as permitted.
- **Blog Author:** Blog module only (out of exam scope).

Registration: Student/Teacher/Institute self-register. Teacher may link to an Institute. Admin can approve/verify Institutes.

SSO: Single Sign-On with JobsXperts Identity Provider; roles mapped via claims.

3. Multi-Tenancy & Subdomains

- Pattern: `{institute-slug}.jobsxperts.com`
- New institute onboarding auto-provisions subdomain and tenant record.
- Data isolation via `tenant_id` in all domain tables (shared DB model) or package support.

4. Question Bank

- Types: **MCQ (single-correct), True/False**
- Metadata: tags, topic, difficulty, explanation, optional media.
- CRUD + Bulk import (CSV/Excel v2).

5. Tests

5.1 Mock Tests

- Availability: anytime.
- Attempts: multiple (limit set per test by Teacher/Institute/Admin/Sub-Admin).
- Results: shown immediately after finish, with answer summary.

5.2 Live Tests

- **Mode A – Windowed:** Start anytime within window; timer gives only remaining time in window.
- **Mode B – Fixed-time:** Starts at exact time; single attempt; no late entry.
- Results: published later by organizers; answers visible after publish.

5.3 Test Settings

- Timers: overall duration + optional section-wise timing.
- Marks: uniform per test (e.g., all questions = 1 or 4 marks).
- Negative marking: custom value per test (e.g., 0, 0.25, 1).
- Randomization: shuffle questions and options on every attempt.
- Resume: allowed on reconnect; autosave every 10s.

6. Scheduling & Access

- Timezone: per-user timezone handling for schedules.
- Access modes:
 - **Apply-only:** students must apply before start; only approved can enter.
 - **Open-unlimited:** anyone can attempt (within window/rules).
 - **Institute-only:** visible and accessible only to that institute's students.
- Optional seat limits (v2); MVP default unlimited unless set.
- Notifications: email (and SMS v2) for schedule, reminders, publish.

7. Attempt Engine

- Secure test runner UI with full-screen prompt and basic tab-switch warning (anti-cheat basics).
- Timer (overall/section) with warning thresholds; autosave every 10s.
- Question navigation: list, flags, next/prev; per-section locks if enabled.
- Submission: auto-submit on time end; manual submit anytime; confirmation modal.

8. Scoring & Results

- Auto-evaluation for MCQ/True-False with uniform marks and configured negative marking.
- Mock: instant result + answer summary (correct options, user choice, explanation).

- Live: results hidden until publish; after publish show scorecard + answers.
- Rankings: global and institute-wise; tie-breaker by finish time, then least incorrect.
- Exports: CSV for results and rankings.

9. Data Model (MVP overview)

Entity	Key Fields
institutes	id, name, slug, subdomain, status, owner_user_id
users	id, name, email, role, institute_id (nullable for global/standalone), timezone
questions	id, institute_id, type (MCQ/TF), stem, options (JSON), correct, tags, difficulty, explanation
tests	id, institute_id, type (mock/live), marks_per_question, negative_mark, duration_total, section_config (JSON), shuffle_q, shuffle_opts
test_sections	id, test_id, title, duration, order, question_ids (JSON)
schedules	id, test_id, mode (window/fixed), start_at, end_at, access (apply/open/institute)
applications	id, schedule_id, user_id, status (applied/approved/rejected)
attempts	id, schedule_id, user_id, started_at, submitted_at, status, score, meta (JSON)
answers	id, attempt_id, question_id, response, is_correct
results	id, attempt_id, total_marks, correct_count, wrong_count, rank_global, rank_institute, published_at

10. APIs (high-level)

- Auth/SSO: OIDC login, callback, token exchange, role claims mapping.
- Question Bank: list/create/update/delete, bulk import.
- Tests: create builder, add sections/questions, settings, preview.
- Schedules: create (window/fixed), access mode, approve applications.
- Attempts: start, autosave, resume, submit.
- Results: compute, publish, rankings, exports.

11. UI Screens (MVP)

Student

- Dashboard (upcoming live, available mock)
- Apply to live test / Start test
- Test runner (timer, nav, autosave)
- Result & answer summary (per rules)

Teacher/Institute

- Question bank (CRUD, tags)
- Test builder (settings, sections)
- Scheduling (window/fixed, access)
- Applications (approve/reject)
- Results & rankings (publish)

12. Security & Compliance

- SSO via OAuth2/OIDC; HTTPS enforced; JWT validation.
- Basic anti-cheat (tab switch warning); logs & audit trail.
- Role-based authorization on all endpoints.

13. Infra & Deployment

- DNS: wildcard `*.jobsxperts.com` to Test Platform.
- Web server: Apache/Nginx vhost route subdomains to the app.
- SSL: wildcard certificate (or on-demand via ACME).
- Queues: for scoring and ranking jobs.

Local Dev: Use `hosts` entries for a few sample subdomains (e.g., `alpha.jobsxperts.local`).

14. Out of Scope (for MVP)

- Advanced proctoring (webcam, ID) – future.
- Payments, coupons – future.
- Item analysis (psychometrics) – basic stats only in v1.

15. Timeline (suggested)

- Week 1: Project setup, SSO, tenancy, core models.
- Week 2: Question bank, test builder, scheduling.
- Week 3: Attempt engine (timer, autosave, resume).
- Week 4: Scoring, results, rankings, exports, QA.

Prepared for JobsXperts • MVP spec generated to guide build. Print this page to PDF (Ctrl+P) for distribution.